

4.3 Further Mechanics

This topic covers momentum and circular motion.

This topic may be studied using applications that relate to, for example, a modern rail transport system.

Students will be assessed on their ability to:	Suggested experiments
73 use the expression $p = mv$	
74 investigate and apply the principle of conservation of linear momentum to problems in one dimension	Use of, for example, light gates and air track to investigate momentum.
75 investigate and relate net force to rate of change of momentum in situations where mass is constant (Newton's second law of motion)	Use of, for example, light gates and air track to investigate change in momentum.
76 derive and use the expression $E_k = p^2/2m$ for the kinetic energy of a non-relativistic particle	
77 analyse and interpret data to calculate the momentum of (non-relativistic) particles and apply the principle of conservation of linear momentum to problems in one and two dimensions	
78 explain and apply the principle of conservation of energy, and determine whether a collision is elastic or inelastic	
79 express angular displacement in radians and in degrees, and convert between those units	
80 explain the concept of angular velocity, and recognise and use the relationships $v = \omega r$ and $T = 2\pi/\omega$	
81 explain that a resultant force (centripetal force) is required to produce and maintain circular motion	

Students will be assessed on their ability to:	Suggested experiments
82 use the expression for centripetal force $F = ma = mv^2/r$ and hence derive and use the expressions for centripetal acceleration $a = v^2/r$ and $a = r\omega^2$.	Investigate the effect of m , v and r of orbit on centripetal force